Truck and Rail Traffic Patterns at U.S. Borders: An Applied Data Science Study

Linh Huynh

Bellevue University

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Prof. Amirfarrokh Iranitalab

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Business Problem

A logistics company wants to understand freight movement at the borders with Canada and Mexico. The company would like to know which border has more activity, how many trucks and rail containers are expected each month, and which ports usually experience busy “peak” months. Without accurate predictions, the company could send too many or too few resources to certain locations. The goal of this project is to build simple and clear models that help managers decide where to invest and how to plan operations.

Background/History

Trade between the United States, Canada, and Mexico is very important for many businesses. Trucks and trains move goods across borders every day. Logistics companies must decide where to send their trucks, how many workers to schedule, and which ports need more space or equipment.

If companies cannot predict traffic well, they may waste money or face delays. This project will use historical border-crossing data to study freight patterns and help a logistics company plan better.

Data source

This project uses the Border Crossing Entry Data from Kaggle. The data comes from U.S. government sources and shows monthly counts of people and vehicles entering the U.S. at land ports.

Important columns include:

- Port name and state
- Border (Canada or Mexico)
- Month and year
- Type of crossing (truck, rail container, passenger vehicle, etc.)
- Number of crossings

The analysis will focus mainly on trucks and rail containers, which are most important for freight companies.

Data Explanation (Data Prep/Data Dictionary/etc.)

Data cleaning focused on preparing the dataset for time-series analysis, forecasting, and port-level comparisons. The Date field was converted to datetime format, and year and month

variables were created to support seasonal and trend analysis. The Value column was converted to numeric, and negative or missing values in critical fields were removed. Records were filtered to include only truck and rail measures, since the project focuses on freight movement. Duplicate observations for the same port, measure, and month were aggregated to ensure accurate totals.

Exploratory data analysis then examined freight patterns across U.S. borders. Total truck and rail traffic were compared between the U.S.-Canada and U.S.-Mexico borders, monthly averages were used to identify seasonal effects, and yearly totals were plotted to show long-term trends. Finally, ports were ranked by total freight volume to determine the busiest locations and highlight where operational planning and forecasting would be most valuable.

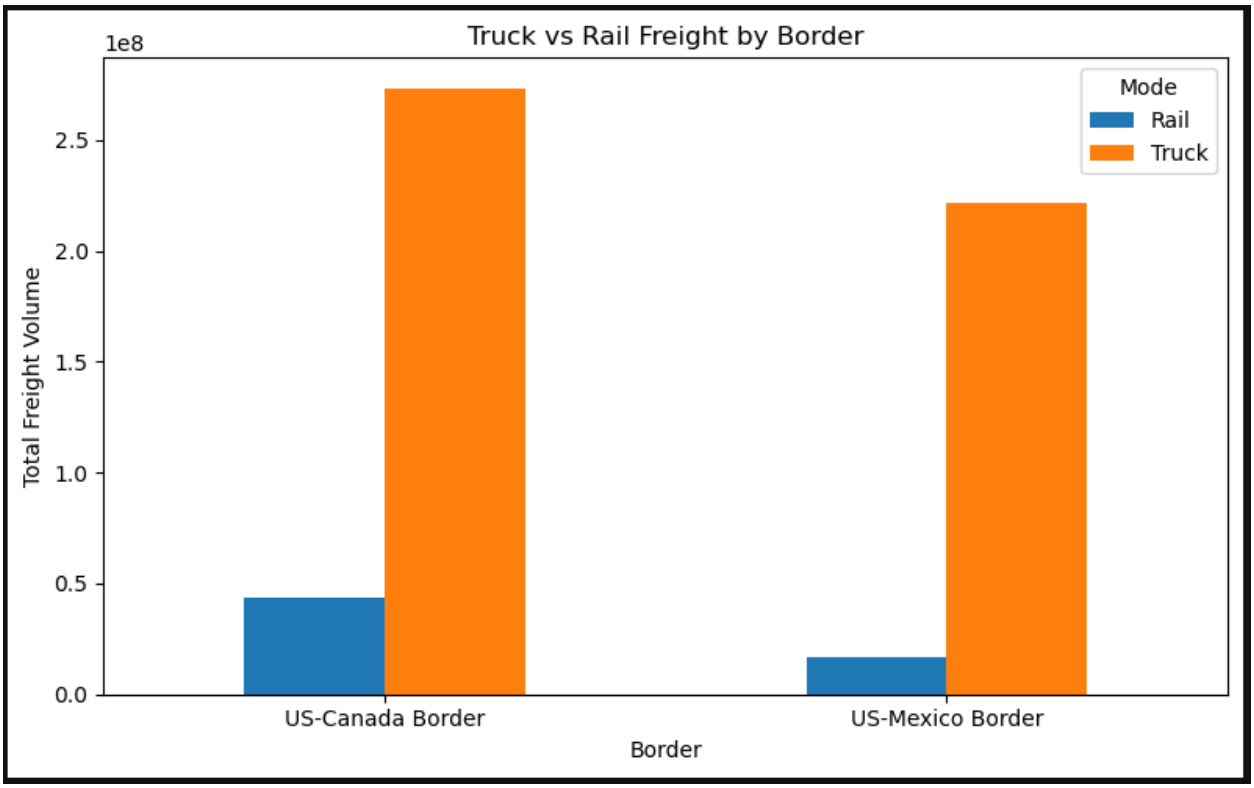
Methods

This project will use several techniques to study border traffic and support business decisions. First, exploratory data analysis will be used to compare total truck and rail activity at each border, examine seasonal patterns by month, plot trends over time, and identify the busiest ports. For forecasting, the project will apply easy-to-understand models such as a seasonal naive method that uses the same month from the previous year, a moving average, and simple linear regression with time and month as predictors. Model accuracy will be compared using MAE and RMSE.

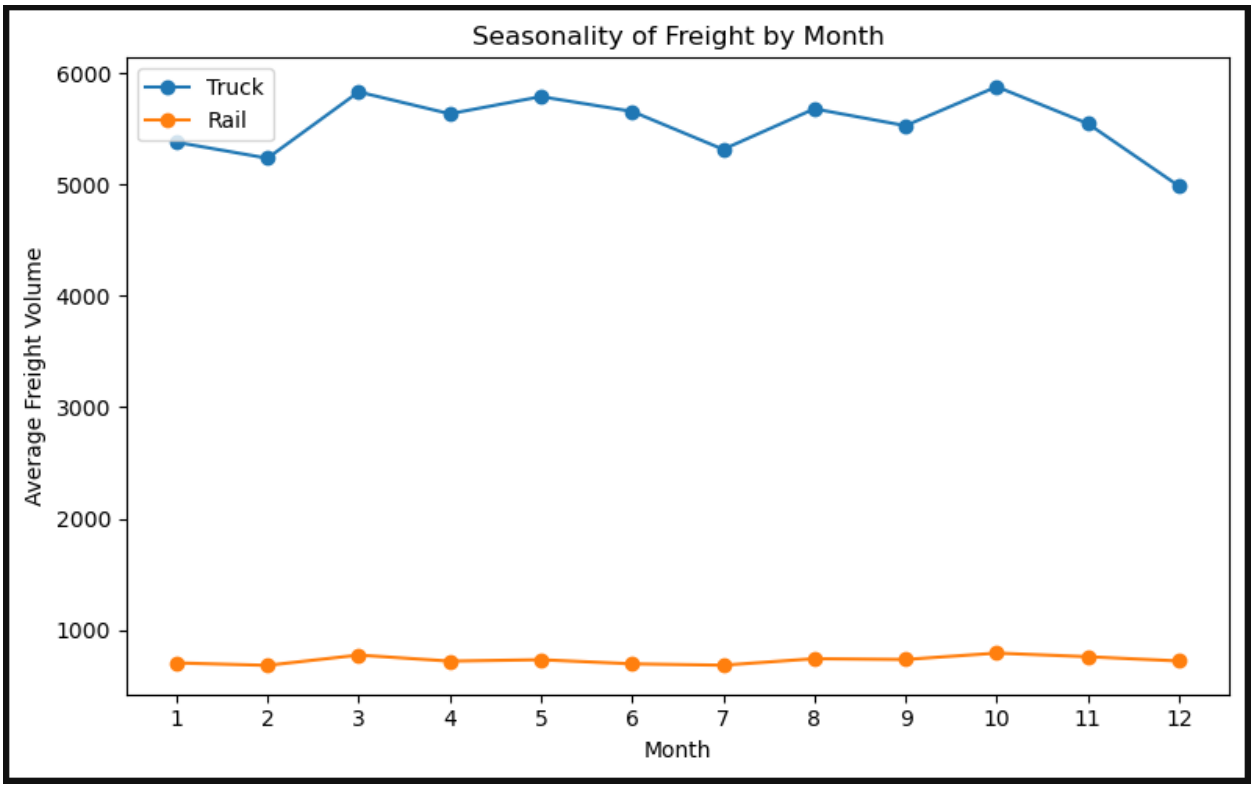
The project will also group ports by activity level. It will calculate the average truck and rail volume for each port and then use K-Means clustering to classify ports into freight-heavy, passenger-focused, or mixed-use groups, with silhouette scores used to check the quality of the clusters.

Analysis

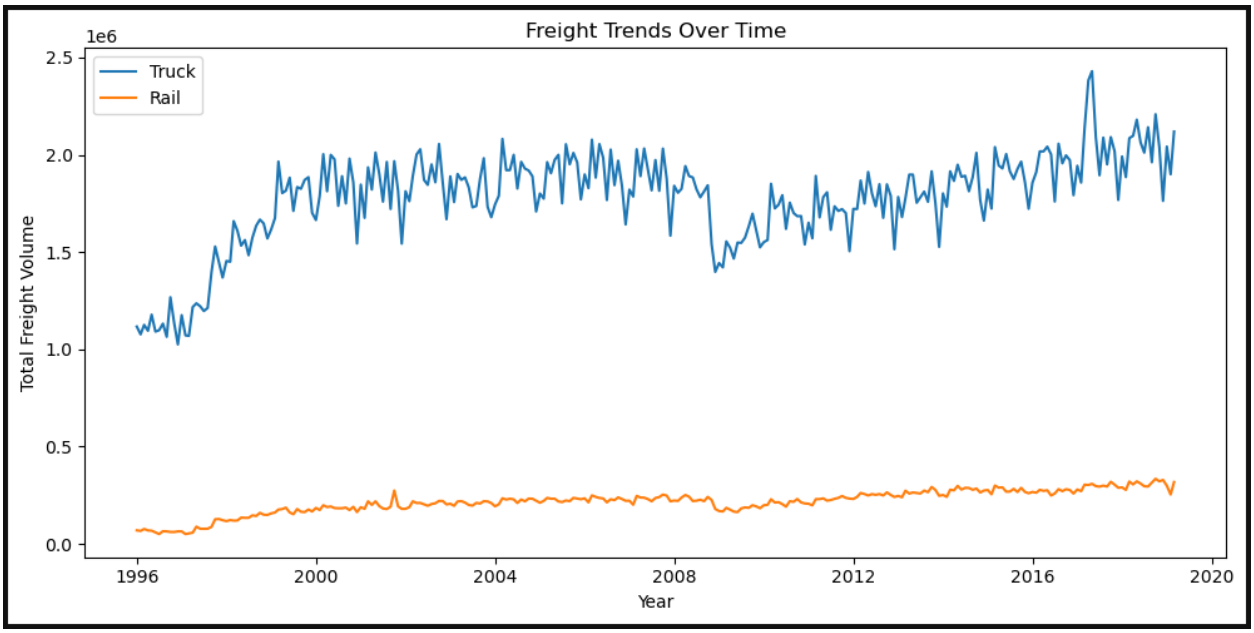
Exploratory Data Analysis



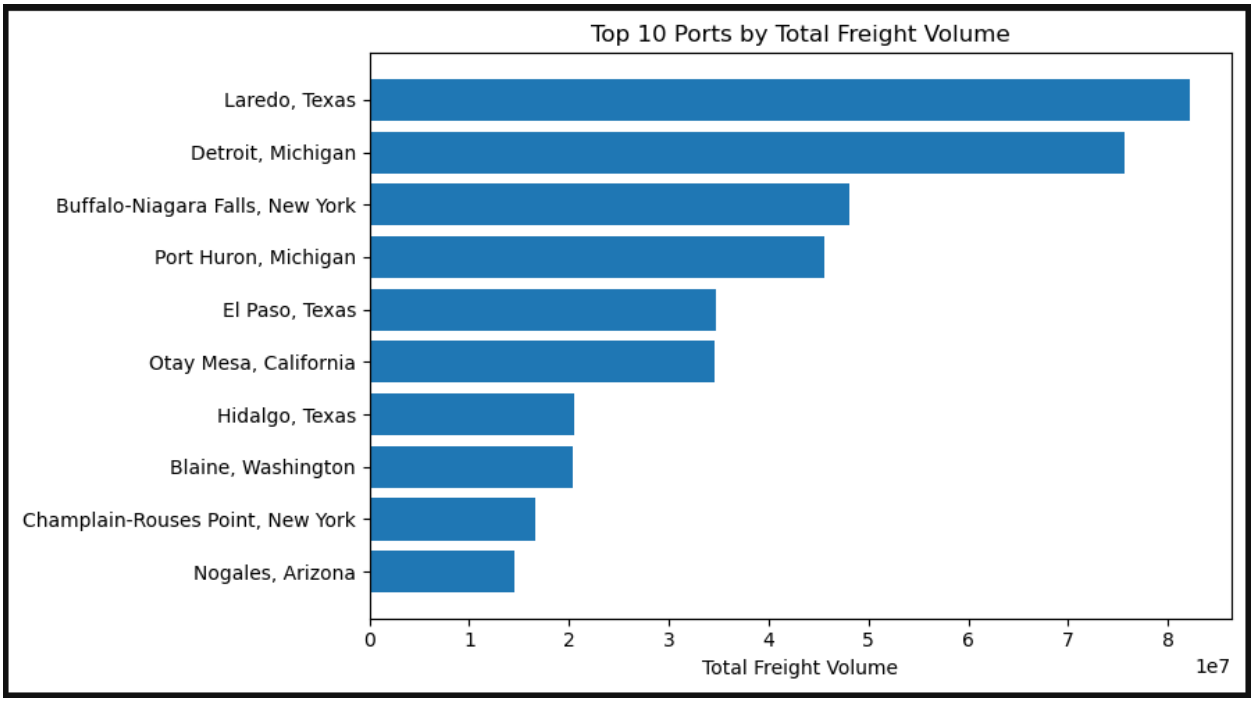
This chart compares total truck and rail freight volume at the U.S.-Canada and U.S.-Mexico borders. Truck traffic is much higher than rail traffic at both borders, showing that trucks are the main way freight crosses U.S. borders. The Canada border handles more rail freight, while the Mexico border relies more heavily on truck transportation.



This chart shows the average monthly freight volume for truck and rail transportation. Both truck and rail traffic follow a seasonal pattern, with higher volumes in spring and fall and lower volumes in summer and December. This suggests that freight activity changes by season and can be planned for staffing and resources in advance.



This chart shows total monthly freight volume for truck and rail transportation over time. Both modes increase over the long term, but there is a noticeable drop around 2008-2009, likely due to economic slowdown. After this period, freight volume recovers and continues to grow, showing long-term growth in cross-border trade.



This chart shows the top 10 ports by total freight volume. Freight traffic is concentrated at a small number of ports, with Laredo, Texas handling the most volume. Several of the busiest ports are located on the U.S.-Mexico border, while major Canada border ports such as Detroit and Buffalo-Niagara Falls also play an important role. These ports are key locations for trade and should be prioritized for planning and investment.

Forecasting

Use simple baseline models to forecast monthly freight volume and compare their accuracy.

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Forecast Accuracy:
Seasonal Naive   | MAE: 119,778 | RMSE: 167,073
Linear Reg.      | MAE: 162,397 | RMSE: 208,013
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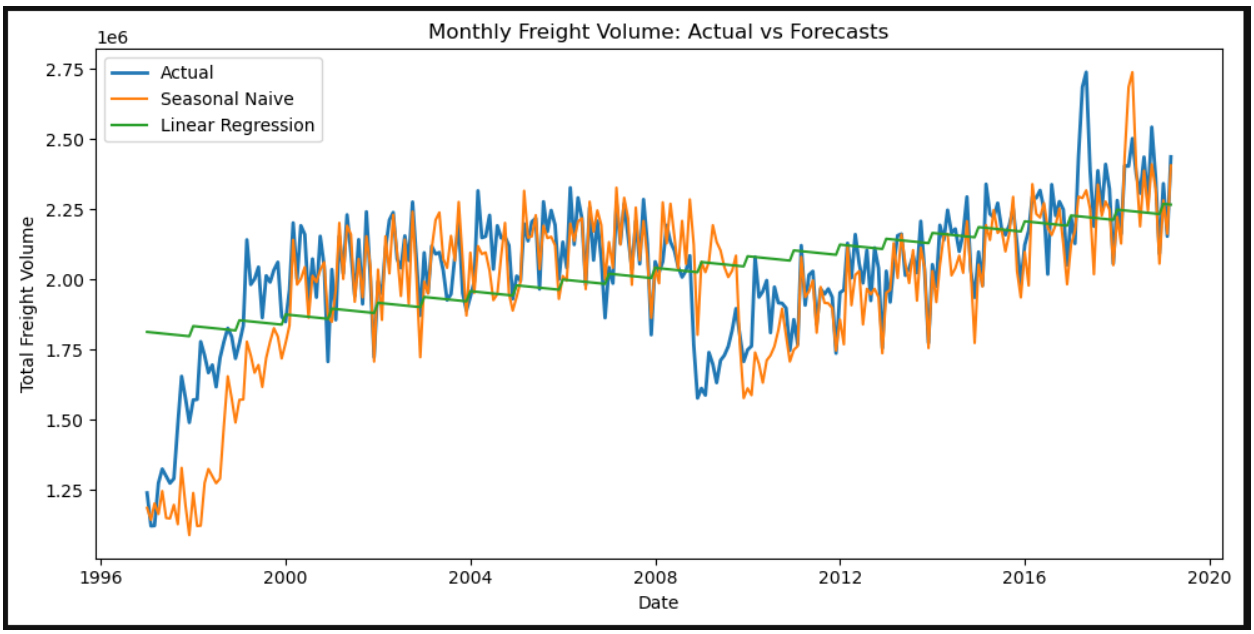
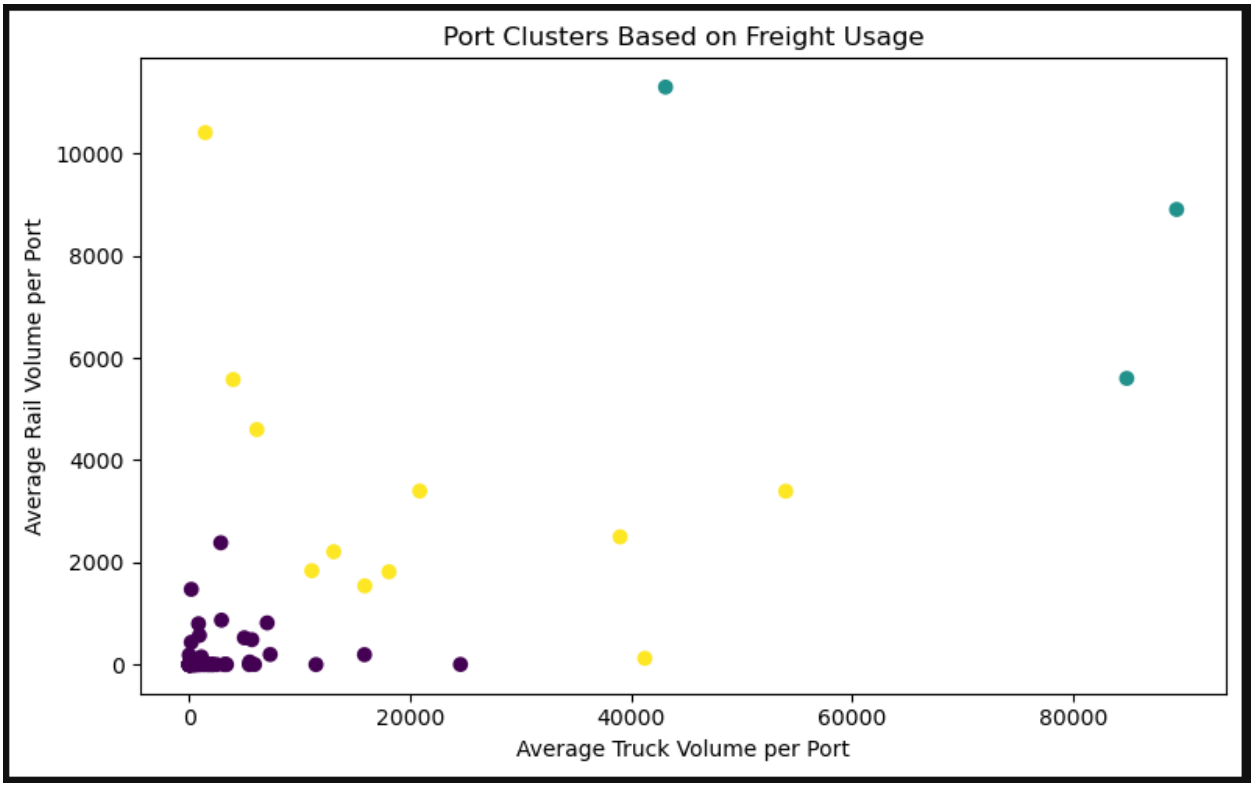


Two forecasting models were compared using MAE and RMSE. The seasonal naive model had lower error than the linear regression model, which shows that monthly freight volume follows strong seasonal patterns. This result suggests that simple baseline models can be effective for short-term freight planning.

Clustering

Group ports based on their average truck and rail freight activity to identify different usage patterns.

Mode	Truck	Rail
Cluster		
0	1494.549095	96.156316
1	72443.438072	8603.974910
2	20443.640925	3400.275243



Ports were grouped into three clusters based on their average truck and rail freight volumes. Cluster 0 represents smaller, truck-focused ports with little rail activity. Cluster 2 includes medium-volume ports that use both truck and rail. Cluster 1 represents major freight hubs with very high activity in both modes. These clusters help identify which ports require different levels of investment and operational focus.

Conclusion

This project analyzed U.S. border-crossing freight data to understand truck and rail traffic patterns, forecast future freight volumes, and group ports based on usage characteristics. Exploratory data analysis showed that truck transportation consistently carries much higher freight volume than rail at both the Canada and Mexico borders. Freight activity follows clear seasonal patterns, with higher volumes in spring and fall, and shows long-term growth with a noticeable drop during the 2008-2009 economic slowdown.

Simple forecasting models were used to predict monthly freight volume. The seasonal naive model outperformed linear regression, indicating that freight traffic is strongly influenced by repeating seasonal patterns rather than a simple linear trend. This result shows that easy-to-interpret baseline models can be effective for short-term planning.

Clustering analysis grouped ports into three distinct usage patterns based on average truck and rail volumes. These clusters clearly separate small truck-focused ports, mixed-use medium-volume ports, and major freight hubs. Together, these findings provide practical insights that can support operational planning, infrastructure investment, and resource allocation at U.S. border ports.

Assumptions

This analysis assumes the dataset accurately reflects monthly truck and rail traffic at U.S. border ports. The Value column is assumed to measure true crossing counts, and aggregated records are treated as correct totals. The final year appears incomplete and will be handled carefully in forecasting. Seasonal patterns observed in the data are assumed to continue, and major economic or policy shocks are treated as special cases rather than regular events.

Limitations

This study has some limitations. The data does not include economic or policy factors that affect freight traffic, which may reduce forecasting accuracy. Some years may be incomplete, especially near the end of the dataset. Clustering results also depend on feature choices and may change with different settings.

Challenges

The main challenges were working with a large dataset, handling incomplete years, dealing with sudden traffic changes over time, and selecting the right number of clusters for port grouping.

Future Uses/Additional Applications

Future work could add economic or weather data to improve forecasts, test more advanced time-series models, and focus on individual ports for staffing and capacity planning.

Recommendations

Business recommendations vary by port cluster based on freight usage patterns. For Cluster 0, which includes truck-focused and smaller ports, investments should focus on improving truck lanes, staffing, and inspection efficiency, since rail usage is very limited. Spending on rail infrastructure at these ports is not recommended because it is unlikely to provide strong returns. For Cluster 2, which represents mixed-use and medium-volume ports, a balanced investment approach across both truck and rail infrastructure is appropriate. Short-term freight forecasts can also be used to adjust staffing and capacity during peak seasons. For Cluster 1, which includes freight-heavy major hubs, these ports should be prioritized for infrastructure upgrades and more advanced forecasting tools because disruptions at these locations have the largest overall impact. Additional staffing and closer monitoring during peak months can help reduce congestion and delays.

Implementation Plan

Update the data regularly, run forecasts to predict traffic and peak months, use port clusters to guide staffing, and adjust plans as conditions change.

Ethical Assessment

The study uses public data with no personal information, and results are meant to support, not replace, human decisions while clearly explaining uncertainty and limits.

Appendix

- Dataset
- Data cleaning code
- Visualization code

10 Questions an audience would ask

1. Why did you focus only on truck and rail traffic instead of passenger crossings?
   Truck and rail data represent freight movement. Passenger crossings are for travel and tourism and do not support freight or logistics decisions.

2. How accurate were your forecasting models, and which one performed best?
   The seasonal naive model was more accurate than linear regression. This shows freight volume follows strong seasonal patterns.

3. How did you handle major events like recessions or border restrictions in the data?
   Major events were kept in the data. They show real changes in freight activity and help explain drops and recoveries.

4. Why does the U.S.-Canada border show higher total traffic than the U.S.-Mexico border?
   The U.S.-Canada border has more ports and strong rail connections. Trade is spread across many crossings.

5. What makes ports like Laredo and Detroit so dominant in freight volume?
   These ports are major trade gateways with strong road and rail links and high industrial activity.

6. How did you decide how many clusters to use for grouping ports?
   Three clusters clearly separate small ports, medium mixed-use ports, and large freight hubs.

7. How reliable are the peak-month predictions for staffing decisions?
   They are reliable for planning because seasonal patterns are consistent but not exact for every year.

8. What additional data would most improve these forecasts in the future?

Economic data, trade policy, fuel prices, weather, and border wait times would improve accuracy.

9. How could border agencies actually use these results in daily operations?
   They can plan staffing, prepare for busy seasons, and focus resources on high-volume ports.

10. What are the biggest risks if decision-makers rely too heavily on these models?
    Unexpected events can change traffic quickly. Models should support decisions, not replace human judgment.

Reference

v11, A. (n.d.). *Border Crossing Entry Data* [Data set]. Kaggle. https://www.kaggle.com/datasets/akhilv11/border-crossing-entry-data